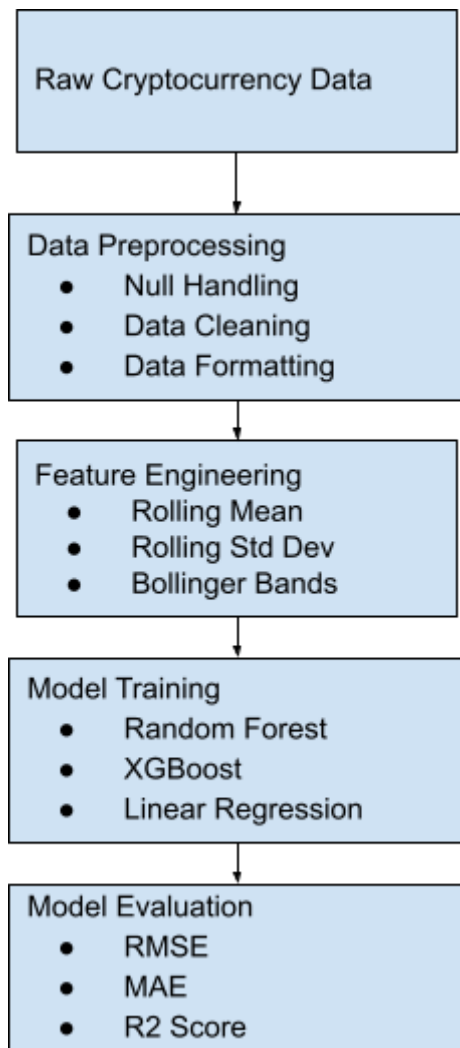


PIPELINE ARCHITECTURE DOCUMENTATION

Pipeline Overview

The project pipeline transforms raw cryptocurrency historical data into actionable volatility predictions through multiple processing stages.

End-to-End Pipeline Architecture



Pipeline Stages Explanation

Stage 1: Data Ingestion

- Load cryptocurrency historical data
- Verify schema consistency
- Validate records

Stage 2: Data Cleaning

- Handle null values
- Remove outliers
- Correct invalid entries

Stage 3: Data Transformation

- Normalize data
- Standardize features
- Convert date formats

Stage 4: Feature Engineering

Generate:

- Volatility measures
- Technical indicators
- Market liquidity indicators

Stage 5: Model Building

Train multiple models and compare performance.

Stage 6: Model Evaluation

Evaluate using:

- RMSE
- MAE
- R² Score

3.4 Machine Learning Pipeline Code Flow

```
# Step 1: Load Data
```

```
load_data()
```

```
# Step 2: Clean Data
```

```
clean_data()
```

```
# Step 3: Feature Engineering  
create_features()
```

```
# Step 4: Split Dataset  
train_test_split()
```

```
# Step 5: Train Model  
train_xgboost()
```

```
# Step 6: Evaluate Model  
evaluate_model()
```

```
# Step 7: Save Model  
save_model()
```

Pipeline Advantages

- Automated workflow
- Better maintainability
- Reusable architecture
- Easy debugging
- Faster experimentation